UNI2 PRO and UNI3 PRO
Modbus RTU communication

manual version 14/05/2026

The device enables remote control and data reading via the **Modbus RTU protocol** . To establish a connection, configure the serial port according to the following parameters:

Connection parameters (RS485)

- **Speed (Baud rate):** 9600 bps
- **Data bits:** 8
- **Parity:** None
- **Stop bits:** 1
- **Device address (Slave ID):** 1 (default)

principles of data reading

Characteristic	Description
Temperature scaling	Values are sent as integers. The result should **be divided by 10** (e.g., a reading of `258` = **25.8°C**).
Addressing	The device uses an offset. Register addresses may differ by 1 depending on the software used.
Response time	Set the *Timeout parameter* to a minimum of **500 ms** (0.5 s).

In the UNI PRO controller, enable Modbus RTU communication, in:
"Settings" [13] → "Common" [13-12] → "Communication" [13-12-3] → "MODBUS RTU" [13-12-3-1]

access requires entering the code "456"

After switching on, it is recommended to briefly press the reset button or turn the controller on again.

Supported Features

Decimal code	Hexadecimal code	Description
3	03	Reading registers
4	04	Reading input registers
6	06	Register entry
16	10	Recording of registers

ATTENTION!

UNI2 PRO does not support CO3 circulation queries

UNI3 PRO does not support queries regarding DHW and Circulation circuits

Address Offset Table (CO1 - CO7)

Circuit	Read Registers (Input)	Holding Registers	Offset
CO1	**30101** – 30102	**40101** – 40130	+0
CO2	**30201** – 30202	**40201** – 40230	+100
CO3	**30301** – 30302	**40301** – 40330	+200
CO4	**30401** – 30402	**40401** – 40430	+300
CO5	**30501** – 30502	**40501** – 40530	+400
CO6	**30601** – 30602	**40601** – 40630	+500
CO7	**30701** – 30702	**40701** – 40730	+600

The query table was created in the following order:

1.	Queries regarding common circulation elements	→	
2.	Inquiries regarding CO1 circulation (or others based on address offset)	→	
3.	Inquiries regarding the DHW circuit	→	
4.	Circulation Inquiries	→	

PARAMETER NAME	DETAILED DESCRIPTION	REGISTER	POSSIBLE DATA OVERWRITE	FUNCTION (HEX)	DATA FORMAT
General condition	**Bit 0:** OK (0) / Alarm (1) **Bit 1:** Boiler relay status K1 (0/1) **Bit 2:** ECO input status (0/1) **Bit 3:** Season Entry (0/1) **Bit 4:** Power sensor 1 error (short circuit) **Bit 5:** Power sensor 1 error (open circuit) **Bit 6:** Power sensor 2 error (short circuit) **Bit 7:** Power sensor 2 error (open circuit) **Bit 8:** External sensor error (short circuit) **Bit 9:** External sensor error (open circuit) **Bit 10:** Buffer pump status (0/1) - *only for UNI2, in UNI3 always 0*	30001	NO	04	**Bit** (16-bit)
Supply temperature sensor 1	-	30002	NO	04	**int16** ($x10$)
Supply temperature sensor 2	-	30003	NO	04	**int16** ($x10$)
Outside temperature sensor	-	30004	NO	04	**int16** ($x10$)
Heating season [10-1]	**0:** Manual **1:** Automatic-date **2:** Automatic-temp **3:** External input	40001	YES	03 and 06	**uint16**
Heating season start date [10-3]	**Byte HI** (older): month (1-12) **Byte LO** (younger): day (1-31)	40002	YES	03 and 06	**2x Byte** (in one register)
Heating season end date [10-4]	**Byte HI** (older): month (1-12) **Byte LO** (younger): day (1-31)	40003	YES	03 and 06	**2x Byte** (in one register)
Heating season start temperature [10-5]	-	40004	YES	03 and 06	**int16** ($x10$)
End of heating season temperature [10-6]	-	40005	YES	03 and 06	**int16** ($x10$)
CO1 circulation status	**Bit 0:** CO1 pump (0: off, 1: on) **Bit 1:** Room input (0: room input open, 1: closed) **Bit 2:** Temperature sensor error (short circuit) **Bit 3:** Temperature sensor error (open)	30101	NO	04	**Bit** (16-bit)
CO1 circulation temperature	-	30102	NO	04	**int16** ($x10$)

PARAMETER NAME	DETAILED DESCRIPTION	REGISTER	POSSIBLE DATA OVERWRITE	FUNCTION (HEX)	DATA FORMAT
Switching on/off the CO1 circuit [13-1-1]	-	40101	NO	03	**Bit**
Set temperature of the CO1 circuit [1-2-1]	-	40102	YES	03 and 06	**int16** ($x10$)
Temperature reported for -20 ° C CO1 cycle [1-2-2] [13-1-5-3]	The set temperature when the weather regulator is switched on and the temperature of -20 ° C is reached on the external sensor	40103	YES	03 and 06	**int16** ($x10$)
Temperature reported for -5 ° C CO1 cycle [1-2-3] [13-1-5-4]	The set temperature when the weather regulator is switched on and the temperature of -5 ° C is reached on the external sensor	40104	YES	03 and 06	**int16** ($x10$)
Temperature reported for +5 ° C CO1 cycle [1-2-4] [13-1-5-5]	The set temperature when the weather regulator is switched on and the temperature at the external sensor reaches +5 ° C	40105	YES	03 and 06	**int16** ($x10$)
Temperature reported for +15 ° C CO1 cycle [1-2-5] [13-1-5-6]	The set temperature when the weather regulator is switched on and the temperature at the external sensor reaches +15 ° C	40106	YES	03 and 06	**int16** ($x10$)
Temporary activation of the CO1 circuit - Time (minutes) [12-1-2]	Adjustable operating time: from 30 to 5940 min (99h) If the value "0" is saved, the timed activation will be disabled.	40110	YES	03 and 06	**uint16**
Temporary activation of the CO1 circuit – Temperature [12-1-3]	-	40111	YES	03 and 06	**int16** ($x10$)
Schedule for CO1 [1-1]	**Byte HI:** Sunday 00:00 - 07:00 **Byte LO:** Sunday 08:00 - 15:00	40120	YES	03 and 10	**2x Byte** (HI/LO)
	Byte HI: Sunday 16:00 - 23:00 **Byte LO:** Monday 00:00 - 07:00	40121	YES		**2x Byte** (HI/LO)
Schedule for CO1 [1-1]	**Byte HI:** Monday 08:00 - 15:00 **Byte LO:** Monday 16:00 - 23:00	40122	YES	03 and 10	**2x Byte** (HI/LO)
	Byte HI: Tuesday 00:00 - 07:00 **Byte LO:** Tuesday 08:00 - 15:00	40123	YES		**2x Byte** (HI/LO)
	Byte HI: Tuesday 16:00 - 23:00 **Byte LO:** Wednesday 00:00 - 07:00	40124	YES		**2x Byte** (HI/LO)

PARAMETER NAME	DETAILED DESCRIPTION	REGISTER	POSSIBLE DATA OVERWRITE	FUNCTION (HEX)	DATA FORMAT
	Byte HI: Wednesday 08:00 - 15:00 **Byte LO:** Wednesday 16:00 - 23:00	40125	YES		**2x Byte** (HI/LO)
	Byte HI: Thursday 00:00 - 07:00 **Byte LO:** Thursday 08:00 - 15:00	40126	YES		**2x Byte** (HI/LO)
	Byte HI: Thursday 16:00 - 23:00 **Byte LO:** Friday 00:00 - 07:00	40127	YES		**2x Byte** (HI/LO)
	Byte HI: Friday 08:00 - 15:00 **Byte LO:** Friday 4:00 PM - 11:00 PM	40128	YES		**2x Byte** (HI/LO)
	Byte HI: Saturday 00:00 - 07:00 **Byte LO:** Saturday 08:00 - 15:00	40129	YES		**2x Byte** (HI/LO)
	Byte HI: Saturday 16:00 - 23:00 **Byte LO:** Unused (always 0x00)	40130	YES		**2x Byte** (HI/LO)
DHW circulation status	**Bit 0:** DHW pump (0: off, 1: on) **Bit 2:** Temperature sensor error (short circuit) **Bit 3:** Temperature sensor error (open circuit)	30801	NO	04	**Bit** (16-bit)
DHW circulation temperature	-	30802	NO	04	**int16** ($x10$)
Turning on/off the DHW circuit [13-8-1]	**Bit 0:** Disabled (0) / Enabled (1)	40801	NO	03	**Bit**
Disinfection [8-4]	**Bit 0:** Disinfection disabled (0) / enabled (1)	40802	YES	03 and 06	**Bit**
Set temperature of the DHW circuit [8-2]	-	40803	YES	03 and 06	**int16** ($x10$)
Temporary activation of the DHW circuit - Time (minutes) [12-8-2]	Adjustable operating time: from 30 to 5940 min (99h) If the value "0" is saved, the timed activation will be disabled.	40804	YES	03 and 06	**uint16**
Temporary activation of the DHW circuit – Temperature [12-8-3]	-	40805	YES	03 and 06	**int16** ($x10$)
Disinfection schedule [8-5]	Disinfection schedule (days of the week) on **7 bits of the lower byte (LO)**	40806	YES	03 and 06	**Bitwise (LO)**
Schedule for DHW [8-1]	**Byte HI:** Sunday 00:00 - 07:00 **Byte LO:** Sunday 08:00 - 15:00	40820	YES	03 and 10	**2x Byte** (HI/LO)

PARAMETER NAME	DETAILED DESCRIPTION	REGISTER	POSSIBLE DATA OVERWRITE	FUNCTION (HEX)	DATA FORMAT
	Byte HI: Sunday 16:00 - 23:00 **Byte LO:** Monday 00:00 - 07:00	40821	YES		**2x Byte** (HI/LO)
	Byte HI: Monday 08:00 - 15:00 **Byte LO:** Monday 16:00 - 23:00	40822	YES		**2x Byte** (HI/LO)
	Byte HI: Tuesday 00:00 - 07:00 **Byte LO:** Tuesday 08:00 - 15:00	40823	YES		**2x Byte** (HI/LO)
Schedule for DHW [8-1]	**Byte HI:** Tuesday 16:00 - 23:00 **Byte LO:** Wednesday 00:00 - 07:00	40824	YES	03 and 10	**2x Byte** (HI/LO)
	Byte HI: Wednesday 08:00 - 15:00 **Byte LO:** Wednesday 16:00 - 23:00	40825	YES		**2x Byte** (HI/LO)
	Byte HI: Thursday 00:00 - 07:00 **Byte LO:** Thursday 08:00 - 15:00	40826	YES		**2x Byte** (HI/LO)
	Byte HI: Thursday 16:00 - 23:00 **Byte LO:** Friday 00:00 - 07:00	40827	YES		**2x Byte** (HI/LO)
	Byte HI: Friday 08:00 - 15:00 **Byte LO:** Friday 4:00 PM - 11:00 PM	40828	YES		**2x Byte** (HI/LO)
	Byte HI: Saturday 00:00 - 07:00 **Byte LO:** Saturday 08:00 - 15:00	40829	YES		**2x Byte** (HI/LO)
	Byte HI: Saturday 16:00 - 23:00 **Byte LO:** Unused (always 0x00)	40830	YES		**2x Byte** (HI/LO)
Circulation State	**Bit 0:** Circulation pump (0: OFF, 1: ON) **Bit 2:** Sensor error (short circuit) **Bit 3:** Sensor error (open circuit)	30901	NO	04	**Bit** (16-bit)
Circulation temperature	-	30902	NO	04	**int16** ($x10$)
Turning the Circulation circuit on/off [13-9-1]	**Bit 0:** Disabled (0) / Enabled (1)	40901	NO	03	**Bit**
Control mode [13-9-2-1]	**Bit 0:** Time Control (0) / Temperature Control (1)	40902	YES	03 and 06	**Bit**
Circulation setpoint temperature [9-2] [13-9-2-2]	-	40903	YES	03 and 06	**int16** ($x10$)
Temporary activation of the Circulation circuit - Time (minutes) [12-9-2]	Adjustable operating time: from 30 to 5940 min (99h) If the value "0" is saved, the timed activation will be disabled.	40910	YES	03 and 06	**uint16**
Schedule for Circulation	**Byte HI:** Sunday 00:00 - 07:00 **Byte LO:** Sunday 08:00 - 15:00	40920	YES	03 and 10	**2x Byte** (HI/LO)

PARAMETER NAME	DETAILED DESCRIPTION	REGISTER	POSSIBLE DATA OVERWRITE	FUNCTION (HEX)	DATA FORMAT
[9-1]	**Byte HI:** Sunday 16:00 - 23:00 **Byte LO:** Monday 00:00 - 07:00	40921	YES		**2x Byte** (HI/LO)
	Byte HI: Monday 08:00 - 15:00 **Byte LO:** Monday 16:00 - 23:00	40922	YES		**2x Byte** (HI/LO)
	Byte HI: Tuesday 00:00 - 07:00 **Byte LO:** Tuesday 08:00 - 15:00	40923	YES		**2x Byte** (HI/LO)
	Byte HI: Tuesday 16:00 - 23:00 **Byte LO:** Wednesday 00:00 - 07:00	40924	YES		**2x Byte** (HI/LO)
	Byte HI: Wednesday 08:00 - 15:00 **Byte LO:** Wednesday 16:00 - 23:00	40925	YES		**2x Byte** (HI/LO)
	Byte HI: Thursday 00:00 - 07:00 **Byte LO:** Thursday 08:00 - 15:00	40926	YES		**2x Byte** (HI/LO)
	Byte HI: Thursday 16:00 - 23:00 **Byte LO:** Friday 00:00 - 07:00	40927	YES		**2x Byte** (HI/LO)
Schedule for Circulation [9-1]	**Byte HI:** Friday 08:00 - 15:00 **Byte LO:** Friday 4:00 PM - 11:00 PM	40928	YES	03 and 10	**2x Byte** (HI/LO)
	Byte HI: Saturday 00:00 - 07:00 **Byte LO:** Saturday 08:00 - 15:00	40929	YES		**2x Byte** (HI/LO)
	Byte HI: Saturday 16:00 - 23:00 **Byte LO:** Unused (always 0x00)	40930	YES		**2x Byte** (HI/LO)

Example:
Checking the general condition where the ECO and Season contacts are closed.

Inquiry:

ID address	Function	Register address	Number of registers	CRC sum
01	04	0000	0001	31CA

Answer:

ID address	Function	Number of data bytes	Register value	CRC sum
01	04	02	000C	B935

Example:
Changing the heating season to automatic based on temperature

Inquiry:

ID address	Function	Register address	Number of registers	CRC sum
01	06	0000	0002	080B

Answer:

ID address	Function	Register address	Number of registers	CRC sum
01	06	9C41	0002	764F

Example:
Temperature reading at the start of the season (Winter)

Inquiry:

ID address	Function	Register address	Number of registers	CRC sum
01	03	0003	0001	740A

Answer:

ID address	Function	Register address	Number of registers	CRC sum
01	3	0002	0032	3991

Example:
Reading the co1 circulation schedule where all hours are active.

Inquiry:

ID address	Function	Register address	Number of registers	CRC sum
01	03	0077	000B	B417

Answer:

ID address	Function	Number of data bytes	Register value	CRC sum
01	03	16	FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF FF 00	AEA7

Example:
Saving a schedule for the CO2 circulation, where it is inactive every day at 6 o'clock.

Inquiry:

ID address	Function	Register address	Number of registers	Number of bytes	Schedule data	CRC sum
01	10	00DB	000B	16	BF FF FF BF FF FF BF FF FF BF FF FF BF FF FF BF FF FF BF FF FF 00	0CF1

Answer:

ID address	Function	Register address	Number of registers	CRC sum
01	10	00DB	000B	F1F5